Calcif Tissue Int (1985) 37:390–394

Calcified Tissue International
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*Laboratory Investigations*

# Liposome-Mediated Calcium Phosphate Formation in Metastable Solutions

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**Summary.** The present study examined the effect of anionic liposomes on precipitate formation in supersaturated calcium phosphate solutions. The liposomes were prepared by dispersing 7:2:1 molar mixtures of phosphatidylcholine, dicetyl phosphate, and cholesterol in buffered aqueous solutions containing 0 or 50 mM inorganic phosphate ($P_I$). Unencapsulated $P_I$ was removed by gel filtration. The liposomes were then suspended in reaction solutions containing 2.25 mM $Ca^{2+}$ and either 0, 1.0, or 1.5 mM $P_I$. All experiments were carried out at 22°C, pH 7.4, and 240 mOsm. External solution $Ca^{2+}$ and $P_I$ losses were found to be appreciable only when the membranes of liposomes containing entrapped $P_I$ were made permeable to $Ca^{2+}$ with the addition to the suspension of the ionophore X-537A. The $Ca^{2+}$ losses, moreover, were up to 3 times as great (1.5 vs 0.5 mM) when accompanied by external $P_I$ losses than in $P_I$-free external solutions where $Ca^{2+}$ alone was involved. Previous studies showed that in this latter situation, decline in external $Ca^{2+}$ concentration was the result of precipitate formation in the aqueous interiors of the liposomes. The present findings suggest that when the external solution phase was metastable, the apparent coupling of large additional $Ca^{2+}$ losses with intraliposomal precipitation was due to secondary precipitation brought about by the seeding action of intraliposomal crystals penetrating into the external solution. The results may explain in part the origin of extravesicular mineral deposits in matrix vesicle calcification.

**Key words:** Apatite — Calcium phosphates — Liposomes — Matrix vesicles.

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Mineral deposition in matrix vesicles appears to be an early stage in the extracellular calcification of many vertebrate hard tissues [1–3]. However, the precise mechanism by which these extracellular, membrane-enclosed bodies initiate mineralization *in vivo* is not fully understood. In a recent synthetic chemical study [4], an effort was made to investigate some of the possible physicochemical aspects of this calcification process by employing artificial phospholipid vesicles (liposomes) as models for native matrix vesicles. It proved possible to carry out *de novo* precipitation reactions within the aqueous interiors of these liposomes by first filling the interior spaces with buffered inorganic orthophosphate ($P_I$) solutions and then making the enclosing membranes permeable to external $Ca^{2+}$ ions with the addition of the lipophilic ion carrier, lasalocid acid (X-537A). The ensuing precipitation was found to proceed in two stages: with the formation of amorphous calcium phosphate (ACP) occurring initially, followed by the hydrolysis of the ACP and its replacement with crystalline apatitelike phases.

The formation of mineral phases in the intraliposomal spaces led also to a disruption in the integrity of the normally stable enclosing membranes. The experiments described in the present report show that when the $Ca^{2+}$-loading of $P_I$-containing liposomes was carried out in metastable Ca-$P_I$ solutions, this breakdown in membrane unity induced by intraspace precipitation resulted in additional precipitation in the surrounding exterior solution space as well.

## Materials and Methods

Anionic liposomes were prepared as previously described [4] by dispersing dried films of 7:2:1 molar mixtures of phosphatidylcholine (PC), dicetyl phosphate (DCP), and cholesterol (chol) (Avanti Polar-lipids, Inc., Birmingham, AL) in PIPES (piperazine-N,N′-bis[2-ethane-sulfonic acid], Sigma Chemical Co., St.

Louis, MO)—buffered solutions (50 mM, pH 7.4) containing either 0 or 50 mM $KH_2PO_4$. The liposome suspensions (15 μmol lipid/ml) were aged undisturbed at 22°C for 20 h, then unencapsulated solution was removed on Sephadex G-50C (Pharmacia Fine Chemicals, Piscataway, NJ) using HEPES (N-2-hydroxyethylpiperazine-N′-2-ethane-sulfonic acid, Sigma Chemical Co., St. Louis, MO)—buffered (10 mM, pH 7.4) NaCl as the eluting medium.

Eluted fractions richest in liposomes (6 μmol lipid/ml, average) were pooled, then mixed with a sufficient volume (0.04 or 0.06 ml/ml) of the $KH_2PO_4$-PIPES solution used in the encapsulation procedure to bring the inorganic phosphate ($P_I$) concentration in the external medium to either 2.0 or 3.0 mM. The liposome suspensions were next mixed with equal volumes of a 50 mM HEPES-buffered NaCl solution (pH 7.4) containing 4.5 mM $CaCl_2$. At this stage, the $Ca^{2+}$, $P_I$, and total buffer concentrations in the external suspension medium were 2.25, 1.0–1.5, and 31 mM respectively. In $P_I$-free control experiments, a KCl-PIPES solution was added to the suspending medium instead of $KH_2PO_4$-PIPES. In other experiments, the suspending medium was also made 0.8 mM in $MgCl_2$ and/or 0.02 mM in $Na_4P_2O_7$ by adding the appropriate amounts of these salts to the $CaCl_2$-HEPES and $KH_2PO_4$-PIPES solutions, respectively, before mixing the latter solutions with the pooled liposome fractions. All solutions were adjusted to 240 mOms with either KCl ($KH_2PO_4$-PIPES solution) or NaCl before using.

Calcium-liposome interactions were initiated by injecting into the liposomal suspension 15 μl/ml of a 0.6 mM ethanolic solution of the carboxylic ionophore X-537A (lasalocid acid, Na salt, Aldrich Chemical Co., Milwaukee, WI). The final ionophore/lipid molar ratio was 0.003. In ionophore-free control experiments, 15 μl of ethanol alone was added to each ml of suspension. In some reactions the $KH_2PO_4$-PIPES was added to an initially $P_I$-free suspending medium at 2 and 24 h after the injection of the X-537A. All reactions were carried out in sealed 5 ml polypropylene tubes incubated at 22 ± 0.1°C.

At regular intervals after injection of the ionophore, portions of the suspension were filtered through 0.22 μm Millex-GS filter assemblies (Millipore Corp., Bedford, MA). Filtrates were analyzed for $Ca^{2+}$ by atomic absorption spectrophotometry and for inorganic phosphate by the UV spectrophotometric method of Murphy and Riley [5]. Although both filtrate $Ca^{2+}$ and $P_I$ were measured, the $Ca^{2+}$ values were used as the primary indicators of reaction-induced changes in the external suspension medium. The $P_I$ values were less reliable for this purpose because of the concomitant leakage of encapsulated $P_I$ into the suspension medium during ionophore-mediated $Ca^{2+}$ loading of the liposomes [4].

Liposome-free reaction solutions identical in electrolyte makeup to the external suspension medium were prepared by mixing the reactants in the same manner as described above but in appropriate volumes of the NaCl eluting buffer alone. In some solutions, only enough $CaCl_2$ was employed to bring the final $Ca^{2+}$ concentration to 1.5 mM. Seeded reactions were initiated in these latter solutions by adding to each ml of solution 0.04 ml of a 24 h-old apatite suspension (1.4 mg crystals/ml) prepared by spontaneous precipitation [4].

## Results

The data in Fig. 1A show that aqueous solutions equivalent in $Ca^{2+}$ and $P_I$ to the most concentrated external suspension medium used in the liposome experiments were stable toward unseeded precipitation for at least 48 h. On the other hand, some loss of filterable $Ca^{2+}$ from the external medium was observed in all liposome suspensions studied over this time period. The smallest losses occurred in suspensions in which the X-537A-primed liposomes contained no encapsulated $P_I$, even in cases where the external suspension medium contained the maximum $P_I$ concentration employed (1.5 mM) (Fig. 1B). Filterable $Ca^{2+}$ losses were similarly small in suspensions where the liposomes also contained entrapped $P_I$ but to which no ionophore was added (Fig. 1C). For the most part, these external solution $Ca^{2+}$ changes (0.1–0.2 mM) could be attributable to the binding of $Ca^{2+}$ to liposome membrane surfaces [4]. The one possible exception was the slight additional $Ca^{2+}$ loss observed in the ionophore-free liposomes at 48 h.

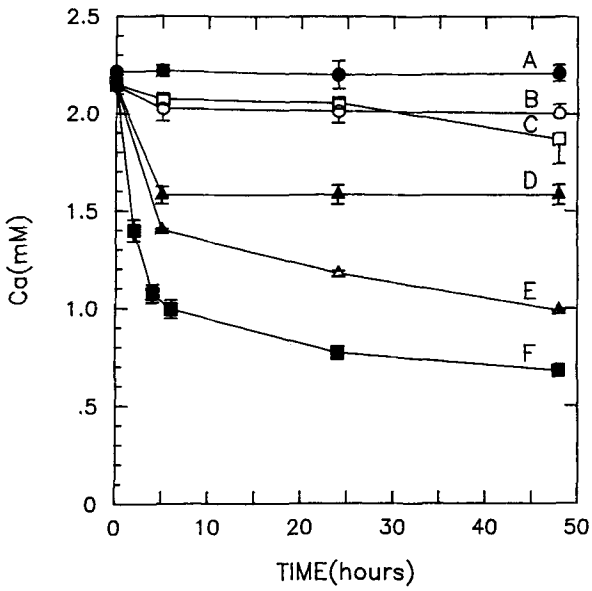


**Fig. 1.** Time-dependent changes in the external solution $Ca^{2+}$ concentration of (*A*) a liposome-free reaction solution initially 2.25 mM in $Ca^{2+}$ and 1.5 mM in $P_I$; (*B,C*) suspensions of X-537A primed, $P_I$-free liposomes and of unprimed, $P_I$-containing liposomes, respectively, in solutions containing 2.25 mM $Ca^{2+}$ and 1.5 mM $P_I$; (*D,E,F*) suspensions of X-537A-primed, $P_I$-containing liposomes in solutions containing 2.25 mM $Ca^{2+}$ and, respectively, 0, 1.0, and 1.5 mM $P_I$. Vertical lines indicate SD of reported values.

Appreciably greater $Ca^{2+}$ losses from the external solution phase occurred when $P_I$-encapsulated liposomes were inoculated with X-537A (Figs. 1D–F). In suspensions containing no external $P_I$, the $Ca^{2+}$ losses stabilized after 5 h at about 0.5 mM (Fig. 1D). As reported previously [4], $Ca^{2+}$ losses in this situation were largely due to calcium phos-

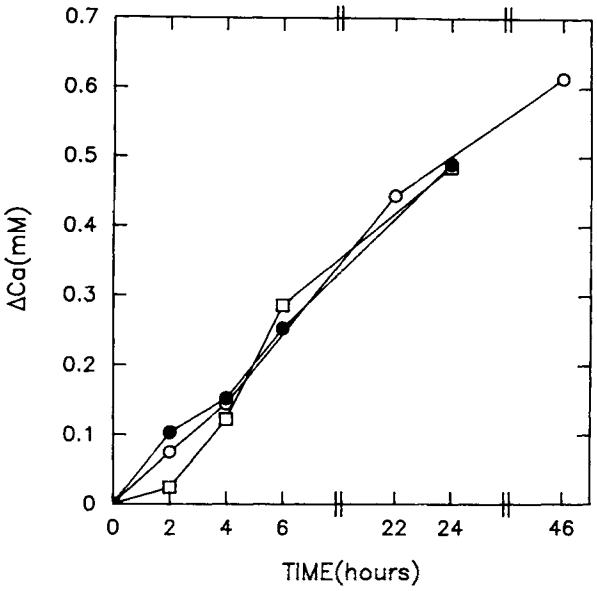


**Fig. 2.** Changes in external solution $Ca^{2+}$ losses (ΔCa) resulting from the addition of 1.5 mM $P_I$ to a X-537A-primed suspension of liposomes containing 50 mM entrapped $P_I$, i.e., ΔCa = Ca losses (1.5 mM external $P_I$)—Ca losses (no external $P_I$). The point at which the $P_I$ was added to the suspension is taken as the origin for each plot. Closed circles (●): $P_I$ added before X-537A, open circles (○): $P_I$ added 2.0 h after X-537A, open squares (□): $P_I$ added 24.0 h after X-537A. External solutions initially contained 2.25 mM $Ca^{2+}$ and 0.8 mM $Mg^{2+}$.

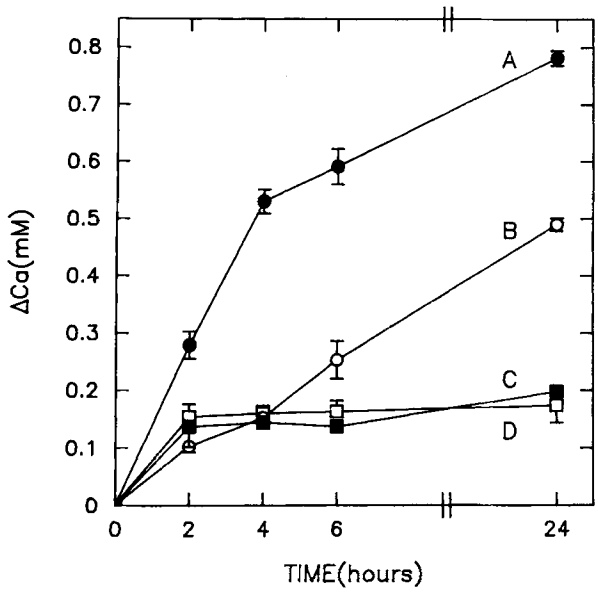


**Fig. 3.** Effect of $MgCl_2$ and $Na_4P_2O_7$ on the changes in $Ca^{2+}$ losses (ΔCa) brought about by the presence of 1.5 mM $P_I$ in the external solution phase of a X-537A-activated suspension of $P_I$-containing liposomes. ΔCa = increase in Ca loss over that which occurred when no external $P_I$ was added. Curve *A:* no Mg, no $P_2O_7$ added; curve *B:* 0.8 mM Mg only added; curve *C:* 0.02 mM $P_2O_7$ only added; curve *D:* 0.8 mM Mg and 0.02 mM $P_2O_7$ added. Initial $Ca^{2+}$ concentration: 2.25 mM. Vertical lines indicate SD of reported values.

**Table 1.** External solution $P_I$ concentrations in liposome suspensions containing 50 mM encapsulated $P_I$ and primed with X-537A

| Reaction time (h) | $P_o$[a] (mM) | $P_{1.5}$[b] (mM) | ΔP[c] (mM) |
|---|---|---|---|
| 5 | 0.31 ± 0.11 SD | 1.41 ± 0.13 SD | 0.40 |
| 24 | 0.38 ± 0.01 SD | 1.13 ± 0.05 SD | 0.76 |
| 48 | 0.38 ± 0.01 SD | 1.13 ± 0.06 SD | 0.75 |

[a] No external solution $P_I$ initially
[b] 1.5 mM external solution $P_I$ initially
[c] $\Delta P = 1.50 + P_o - P_{1.5}$

phate precipitation in the interior aqueous compartments of the liposomes. The observation that further losses did not occur beyond 5 h suggests that all encapsulated $P_I$ had either combined with $Ca^{2+}$ or had leaked out by this time. On the other hand, suspensions that contained 1.0 or 1.5 mM $P_I$ in the external medium continuously decreased in filterable $Ca^{2+}$ (Figs. 1E,F) until at 48 h the maximum measured $Ca^{2+}$ loss was 3 times that observed in the $P_I$-free suspension. However, commensurate $Ca^{2+}$ decreases could be achieved in this latter suspension by adding $P_I$ to the external medium as late as 24 h after priming with X-537A (Fig. 2). These delayed losses appear not to have resulted from extraneous precipitation induced directly by the $P_I$ addition itself. Adding in a similar fashion the $P_I$ after the $Ca^{2+}$ to an equivalent suspension containing no encapsulated $P_I$ did not result in measurable $Ca^{2+}$ changes.

The data in Table 1 show that external solution $P_I$, if present initially, also appeared to decline in concentration after priming $P_I$-encapsulated liposomes with X-537A. However, the leakage of encapsulated $P_I$ (Table 1, $P_o$ column) into the external solution phase makes quantitation of this loss less precise than that for $Ca^{2+}$.

Adding $Mg^{2+}$ and pyrophosphate ions to X-537A-primed liposome suspensions containing both encapsulated and external $P_I$ greatly decreased the rate at which filterable $Ca^{2+}$ was removed (Fig. 3). These ions, however, did not affect the amount of $Ca^{2+}$ taken up by the $P_I$-liposomes in suspensions that contained no external $P_I$. The addition of 0.8 mM $Mg^{2+}$ slowed the $Ca^{2+}$ loss from external solutions containing 1.5 mM $P_I$ to a rate comparable to that observed in solutions with only 1.0 mM $P_I$. As little as 0.02 mM pyrophosphate anions, on the other hand, totally inhibited further $Ca^{2+}$ removal

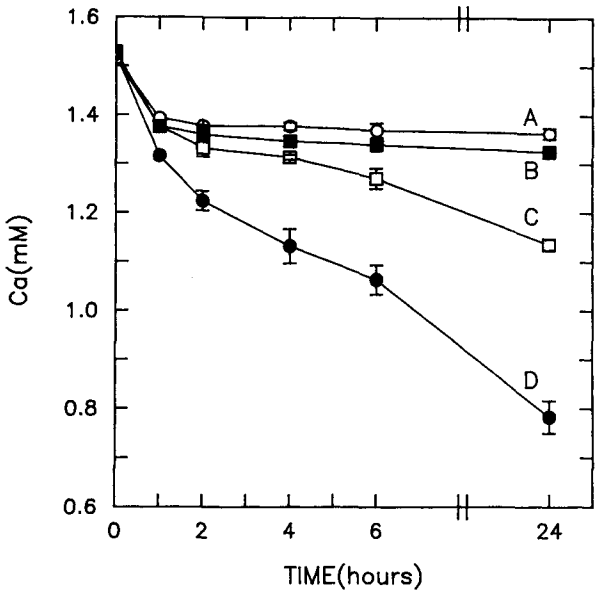


**Fig. 4.** Time-dependent $Ca^{2+}$ changes in liposome-free reaction solutions seeded with 1.4 mg apatite/ml. All solutions were initially 1.5 mM in $Ca^{2+}$ and 1.5 mM in $P_I$. In addition, solution *A* contained 0.8 mM $Mg^{2+}$ and 0.02 mM $P_2O_7$, solution *B* contained 0.02 mM $P_2O_7$, solution *C* contained 0.8 mM $Mg^{2+}$. Solution *D* contained neither of these two reactants. Vertical lines indicate SD of reported values.

after 2 h. The data curves in Fig. 4 show that similar inhibiting effects could be achieved with $Mg^{2+}$ and pyrophosphate ions when added to stable liposome-free supersaturated solutions seeded with apatite crystals.

## Discussion

The data presented show that anionic 7PC: 2DCP:1chol liposomes have a much stronger destabilizing influence on solutions containing both $Ca^{2+}$ and $P_I$ than on solutions containing $Ca^{2+}$ alone. This difference in reactivity was evident, however, only when the aqueous interiors of the liposomes both contained $P_I$ and were made permeable to $Ca^{2+}$ influx with X-537A, the two conditions previously found to be necessary for intraliposomal calcium phosphate formation [4]. When either X-537A or encapsulated $P_I$, or both, was excluded from the experimental protocol, the $Ca^{2+}$ change in the external aqueous solution was the same whether or not this solution was rendered metastable with $P_I$. These findings suggest that the additional $Ca^{2+}$ losses observed in metastable external solutions were tied to calcium phosphate formation inside the liposomes. The further finding that these additional losses were readily induced even when the external $P_I$ was added as late as 24 h after intraliposomal $Ca^{2+}$ loading commenced suggests that these losses were secondary to intraliposomal precipitate formation.

Since decreases in external solution $P_I$ accompanied the secondary solution $Ca^{2+}$ losses, the inference can be drawn that these changes were the result of additional calcium phosphate formation. It is unlikely, however, that this additional precipitation occurred inside the liposomes proper. Rather, the combined effects of low intrinsic membrane permeability to $P_I$ and high intraliposomal $P_I$ concentrations most probably restricted the precipitate to the extraliposomal fluid space. The anionic liposomes used in this study are normally quite resistant to transmembrane $P_I$ movement, even when primed with X-537A [4]. Therefore, only negligible external $P_I$ would be expected to accompany X-537A-mediated $Ca^{2+}$ transport *per se* across intact membranes into the aqueous interiors of the liposomes. But even an increase in $P_I$ permeability, such as observed during intraliposomal precipitate development [4], would not necessarily result in a net uptake of external $P_I$ by the liposome. To the contrary, the large initial $P_I$(encapsulated)/$P_I$(external) ratio ($\geqslant$33) would favor a net efflux of $P_I$ from the liposomal interior and an elevation in external $P_I$ concentration. In fact, results previously reported [4] on the leakage of encapsulated $P_I$ into an initially $P_I$-free external solution phase during intraliposomal precipitation showed that the amount of $P_I$ released was appreciable, approaching nearly 1/2 of that initially entrapped. In the present experiments, $P_I$ released in this way could have raised the external solution $P_I$ concentration in the metastable solutions by as much as 0.4 mM (Table 1).

In itself, the elevation in external $P_I$ brought about by encapsulated $P_I$ leakage during intraspace precipitation was probably not sufficient to induce extraliposomal precipitation in the metastable solutions used in the present study, although this leakage may have contributed to subsequent mineral development. In reactions where 1.0 mM $P_I$ was initially present in the external solution, the increase in $P_I$ concentration was still well below the stability limit for the solutions used in this study. Also, as shown previously [4], most of the $P_I$ leakage occurred after the intraliposomal solids had formed so that, even where 1.5 mM $P_I$ was present initially in the external solution, the decline in external $Ca^{2+}$ associated with intraliposomal precipitation would probably offset the destabilizing effect of the $P_I$ leakage.

On the other hand, leakage of encapsulated $P_I$ suggests that intraliposomal precipitation may have disrupted the integrity of the liposomal membranes [4]. Such breakdown in membrane structure could have resulted in exposure of the internally formed solids to the external solution. Once penetrated, the solids could become substrates for seeding the external solution. Additional evidence that the intraliposomal precipitate induced external precipitation was the finding that the retarding effects of $Mg^{2+}$ and $P_2O_7$ on the kinetics of the secondary $Ca^{2+}$ losses were similar to those observed in metastable solutions seeded with apatite crystals.

Several studies [6–8] indicate that the first apatite crystals seen in association with matrix vesicles originate in the aqueous interiors of these cell-derived bodies. However, the bulk of the apatite crystals which subsequently form occur outside the vesicles. These extravesicular crystals appear to cluster initially in the fluid space immediately adjacent to the vesicle surface, but later they proliferate outward to form coalescing, nodular crystalline masses [9–11]. This progressive spherulitic growth away from the matrix vesicles suggests that the outer surface of the vesicle membrane provided sites for the initial nucleation of these crystal clusters. The formation of phospholipid-Ca-$P_I$ complexes, which are capable of inducing precipitation in metastable solutions *in vitro* [12, 13], may be one way by which such sites could be activated. An alternative possibility [3] is that it is the mineral already present inside the vesicles that initiates extravesicular mineral formation. Our results support this latter hypothesis and suggest further that one way by which these crystals could fulfill this function is to physically breach the bilayer membrane and then either continue to grow or seed directly new crystals in the surrounding fluid space.

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